Firearm Violence and Legislation in the United States

Data Visualization Report

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Data Visualization — U-tad

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Contents

1 Introduction

This project looks at how firearm laws relate to gun violence across U.S. states from 1976 to 2023. Estimated household gun ownership is included as background context, but it is not the main focus. Countries like Finland and Switzerland show that high ownership does not automatically mean high violence. The central question is whether the type and strength of a state's firearm laws are associated with lower homicide rates, and whether legislative changes correspond to visible shifts over time.

1.1 Methodology

The project followed these steps in order:

1. Select the datasets and plan the join strategy
2. Explore the data outside Tableau
3. Define the research questions
4. Choose chart types suited to each question
5. Connect the data in Tableau and build calculated fields
6. Add interactive features: parameters, sets, filters, actions, and tooltips
7. Build dashboards
8. Use AI for repetitive or mechanical tasks only (see Section 1.3)

1.2 Sources

The project draws on the following sources:

- Official dataset documentation and codebooks[1,2]
- Harvard Dataverse and Tufts CTSI metadata[3,2]
- Academic literature on firearm ownership proxies[4,3]
- Academic literature on the effects of gun laws[5,6,7]
- Tableau official documentation[8]
- Lecture notebooks and course materials
- AI assistance, declared in Section 1.3

All sources appear in the bibliography. AI use is documented with the model name and the exact prompt at each point of use.

1.3 AI Usage Declaration

AI was used only for repetitive, mechanical, or formatting tasks. Every output was reviewed before being included in the project. The analytical conclusions, Tableau visualizations, and interpretation of results are entirely my own. The model used throughout was **Claude Sonnet 4.6**, accessed via the Perplexity AI assistant. Prompts and short response summaries are noted at each point of use, and Table 1 lists them in one place.

Task	Model	Section
Dataset research and pairing strategy	Claude Sonnet 4.6	2
State-abbreviation calculated field	Claude Sonnet 4.6	4.1
Document labelling and cross-referencing	Claude Sonnet 4.6	1.3.1
Report structuring and LaTeX formatting	Claude Sonnet 4.6	1.3.2
Supplementary chart generation	Claude Sonnet 4.6	6

Table 1: AI usage index

1.3.1 Document Labelling and Cross-Referencing

Applying `\label` and `\ref` tags consistently across a long document is tedious and error-prone. It does not require any analytical judgement, so it was a reasonable task to delegate to AI.

> **Prompt:** *Add labels to questions Q1, Q2 and Q3 in the document and reference those labels whenever those questions are referenced in the document.*

> **Response summary:** Added `\label{q1}`, `\label{q2}`, and `\label{q3}` to the question headings, then inserted matching cross-references throughout the question list, conclusions, and visualization summary table.

1.3.2 Report Structuring and LaTeX Formatting

AI was also used for page layout and figure placement. These are formatting details with no effect on the content or analysis.

> **Prompt:** *For visualizations, make it so each of them is on a different page alongside with their text section, so page breaks are consistent and it doesn't break in weird places.*

> **Response summary:** Added `\usepackage{float}`, inserted `\newpage` before each visualization subsection, and switched figure placement from `[h]` to `[H]` so figures appear exactly where they are defined.

2 Datasets

Two public datasets were used. Both are available online and share a state-year structure, making them straightforward to combine in Tableau. The first dataset was chosen independently, and AI helped identify the second one.

> **Prompt:** *I already have a state-year level dataset on estimated household gun ownership in the US. I need a second dataset that contains information about firearm legislation on a state-year basis so I can join them easily. What publicly available datasets would you recommend?*

> **Response summary:** Recommended the Tufts CTSI State Firearm Law Database because it tracks firearm laws by state and year, matches the grain of the ownership proxy dataset, and is widely cited in public health research.

2.1 Dataset 1: Firearm Suicide Proxy for Household Gun Ownership, 1949–2023

The first dataset[1] provides a state-year estimate of household gun ownership across the United States. Since direct ownership data is not publicly available at the state level, the dataset uses the firearm suicide proxy method: the share of suicides committed with a firearm (FSS) serves as an indirect measure of gun prevalence. It is hosted on Harvard Dataverse.

- **Source**: Harvard Dataverse, Kang et al.[3]
- **Grain**: One row per U.S. state per year
- **Key columns**: State, Year, FSS, firearm suicides, total suicides, firearm homicide rate, total homicides, population
- **Date span**: 1949–2023 (75 years, 50 states + D.C.)
- **Format**: Comma-separated values (`.csv`)

FSS is not a literal count of guns. It is a proxy measure with a long track record in public health and policy research.[4] In this project it appears only as background context in the scatter plot. The primary outcome variable throughout is firearm homicide rate.

2.2 Dataset 2: State Firearm Law Database, 1976–2024

The second dataset[2] is maintained by the State Firearm Law Research Group at Tufts University. It records whether each of 72 firearm law provisions was in effect in each U.S. state for every year from 1976 to 2024.

- **Source**: Tufts CTSI, State Firearm Law Database[2]
- **Grain**: One row per U.S. state per year
- **Key columns**: State, Year, 72 binary law indicators (e.g. permit requirements, background checks, waiting periods, stand-your-ground laws, child access prevention)
- **Date span**: 1976–2024
- **Format**: Excel (`.xlsx`)

This dataset was chosen because it gives a year-by-year picture of state legislation over nearly five decades, making it well-suited for studying how law changes relate to homicide trends. It is widely cited in academic public health research.[2]

2.3 Join Strategy

The two datasets were connected in Tableau using a **relationship** on `state` and `year`, configured as **many-to-many** with **some records match** on both sides. Both tables share the same state-year grain, so many-to-many is the correct setting. The "some records match" option prevents Tableau from dropping rows where a key exists in one table but not the other. Tableau aggregates each table independently before combining results, which avoids row duplication. The overlapping date range gives a final analysis window of **1976–2023**.

2.4 Analysis

Both files were inspected before connecting them in Tableau to check structure, labels, and any potential issues. Key findings:

- Both datasets use identical full state names, so joining on state works without any transformation
- The ownership proxy dataset already contains pre-calculated rate columns
- The law dataset includes 72 binary law columns and a total law count field (`lawtotal`)
- The overlapping year range is 1976–2023

2.5 Processing

Both files were loaded into Tableau without any external pre-processing. No cleaning, renaming, or deduplication was needed. The only change was to exclude the District of Columbia, which appears in the ownership dataset but not in the law dataset. All charts are also filtered to 1976–2023, the period shared by both sources.

2.6 Research Questions

Three questions were defined after exploring the data:

1. Is a state's firearm homicide rate associated with the number and type of firearm laws it has in place?[2.6.1]
2. How have firearm homicide rates changed over time across states grouped by the quantity or type of laws they have passed?[2.6.2]
3. For a specific state, does an increase in firearm laws correspond to a lower homicide rate and a better ranking relative to other states?[2.6.3]

2.6.1 Question 1: Homicide Rates and the Law Landscape

Q1 is the starting point. Before examining trends over time, the aim is to check whether a pattern already exists in the cross-sectional data: do states with more laws, or different types of laws, tend to have lower firearm homicide rates? FSS is included only as background context on estimated ownership.

2.6.2 Question 2: Homicide Trends by Law Profile Over Time

Q2 extends Q1 by adding the time dimension. States are grouped by their law profile, either by quantity tier or by law type category (see Section 3.2), and their homicide rate trends are compared across the full 1976–2023 period.

2.6.3 Question 3: State-Level Law History and Homicide Standing

Q3 brings the analysis down to a single state. The viewer can pick any state and see how its law history compares with its firearm homicide ranking relative to the rest of the country. This connects the broad national patterns from Q1 and Q2 to something more concrete and personally relevant.

3 Visualization Prep

Several design decisions were made before building the charts in Tableau to ensure consistency across the project.

3.1 Storytelling

The project is structured as four acts. Act 1 sets the scene with a geographic and cross-sectional view of homicide rates and the law landscape. Act 2 adds time, comparing how rates have evolved across the law categories in Section 3.2. Act 3 zooms into individual states to connect each one's law history to its homicide standing. Act 4 compares the two law groups, Popular and Evidence-Based, to see which one shows a stronger association with lower homicide rates.

3.2 Law Categorization

Two grouping schemes are used across the visualizations. A shared parameter lets the viewer switch between them at any point.

By quantity, states are divided into three tiers based on their total law count (`lawtotal`):

- **Low**: 10 or fewer laws
- **Medium**: 11 to 25 laws
- **High**: more than 25 laws

By type, rather than using all 72 laws, the analysis uses a smaller subset: the three most commonly adopted laws ("Popular") and the three with the strongest evidence base in the research literature ("Evidence-Based"). The three Popular laws are `permitconcealed` (concealed-carry permit requirement), `waiting` (waiting period for firearm purchase), and `statechecks` (state-level background check). The three Evidence-Based laws are `universal` (universal background checks), `gvro` (gun-violence restraining order), and `assault` (assault-weapon restrictions). States fall into one of four categories depending on how many laws from each group they have adopted:

- **Weak**: fewer than two Popular laws and fewer than two Evidence-Based laws
- **Popular**: at least two Popular laws, but fewer than two Evidence-Based laws
- **Evidence**: at least two Evidence-Based laws, but fewer than two Popular laws
- **Both**: at least two laws from each group

3.3 Colors and Themes

A single consistent color palette is used throughout all visualizations, so the same law profile colors appear identically in every chart that groups states by law profile. The choropleth map follows the same palette but adds an orange gradient to express severity: lighter shades indicate lower firearm homicide rates and darker shades indicate higher ones, extending the base scheme to convey continuous magnitude.

- **Neutral colors** throughout to avoid implying a political stance. Grey marks states with weaker law profiles or little legislative change.
- **Orange severity gradient** on the choropleth map, where darker means a higher firearm homicide rate.
- **Minimal gridlines** so the data marks stay in focus rather than the chart frame.

4 Visualizations

All charts follow the color scheme from Section 3.3 and the law categories from Section 3.2. Features introduced in earlier charts are reused and referenced where they appear again.

4.1 Q1 – Homicide Rates and the Law Landscape

Two charts answer Q1[2.6.1]. The first is a **choropleth map** (Figure 1) showing the firearm homicide rate by state for a selected year. Darker states have higher rates. A **year parameter** (`P Year`) controls the view, and a **calculated field** (`CF StateAbbreviation`) displays two-letter state abbreviations on each mark. This is the first chart to introduce the year parameter and the state abbreviation calculated field; both are reused across subsequent charts and dashboards.

Writing all 50 CASE clauses by hand would have been error-prone with no analytical value, so this field was generated with AI (Claude Sonnet 4.6; see Section 1.3).

> **Prompt:** *Generate a Tableau calculated field using a CASE statement that maps each U.S. state's full name to its two-letter abbreviation for all 50 states. Only output the code.*

> **Response:**

```
CASE [State]
WHEN "Alabama" THEN "AL"
WHEN "Alaska"  THEN "AK"
...
WHEN "Wisconsin" THEN "WI"
WHEN "Wyoming"   THEN "WY"
END
```

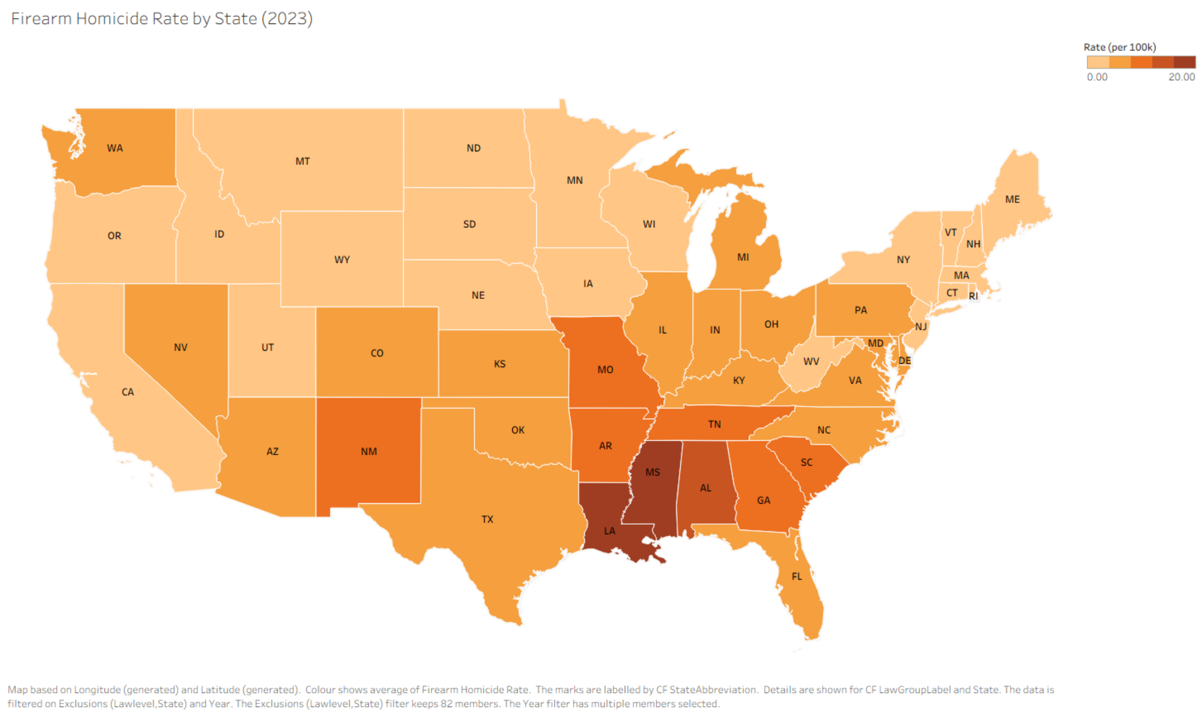

Figure 1: Q1 – Firearm Homicide Rate by State (Choropleth Map)

Insights:

- States with fewer laws tend to have higher firearm homicide rates, visible across different years via the `P Year` parameter.
- A clear geographic division emerges: states with lower rates cluster in the north and northeast, while those with higher rates concentrate in the south. This pattern correlates with the type and quantity of legislation in each region, with southern states generally maintaining fewer restrictive firearm

laws, likely a reflection of the different legislative approaches on either side of this geographic divide.

The second chart (Figure 2) is a **scatter plot** where each point is a state. The X axis shows estimated gun ownership (FSS), the Y axis shows total law count (`lawtotal`), and color encodes the selected law profile. A **parameter** (`P LawCategorization`, first introduced here) switches between the quantity-based and type-based groupings from Section 3.2. This parameter is reused in every subsequent chart that groups states by law profile.

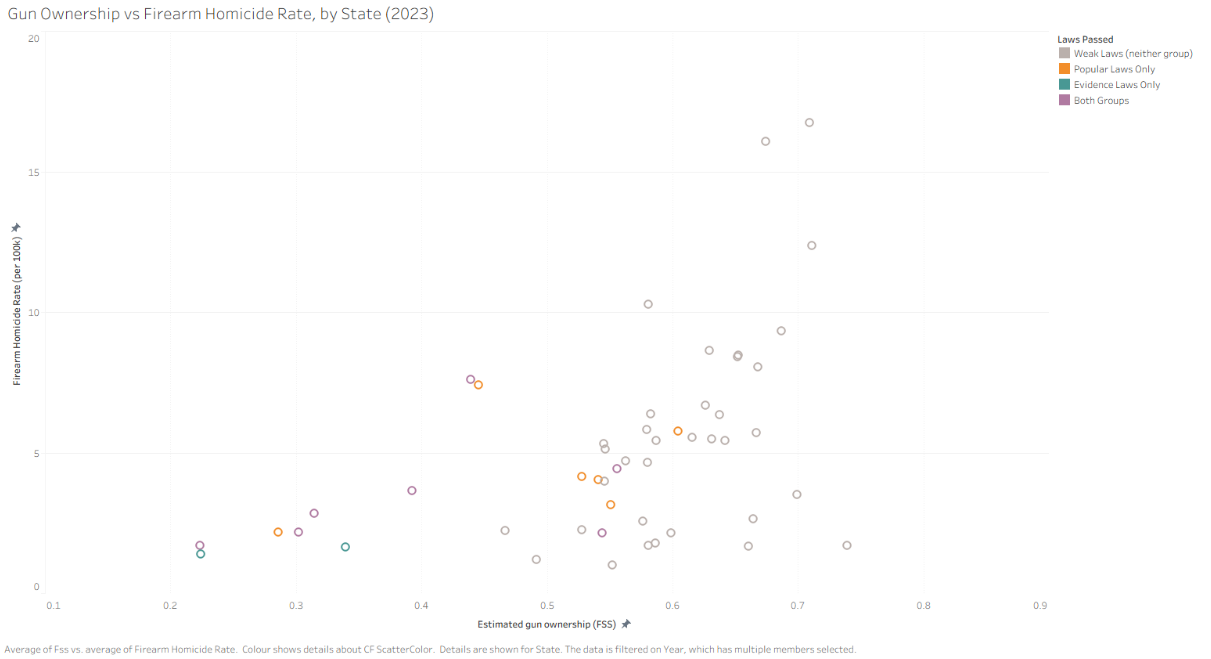

Figure 2: Q1 – Law Count vs. Estimated Gun Ownership (FSS), colored by Law Profile (Scatter Plot)

Insights:

- States with fewer laws tend to cluster at higher firearm homicide rates.
- FSS provides ownership context, but law count and law type are the main analytical lenses.

4.2 Q2 – Homicide Trends by Law Profile Over Time

Q2[2.6.2] uses an **interactive line chart** (Figure 3). The X axis covers 1976–2023, the Y axis shows firearm homicide rate, and each line represents a group of states. The grouping is controlled by the same `P LawCategorization` parameter introduced in Figure 2. In quantity mode the groups are Low, Medium, and High, and in type mode they are Weak, Popular, Evidence, and Both. A vertical **reference line** marks notable legislative periods.

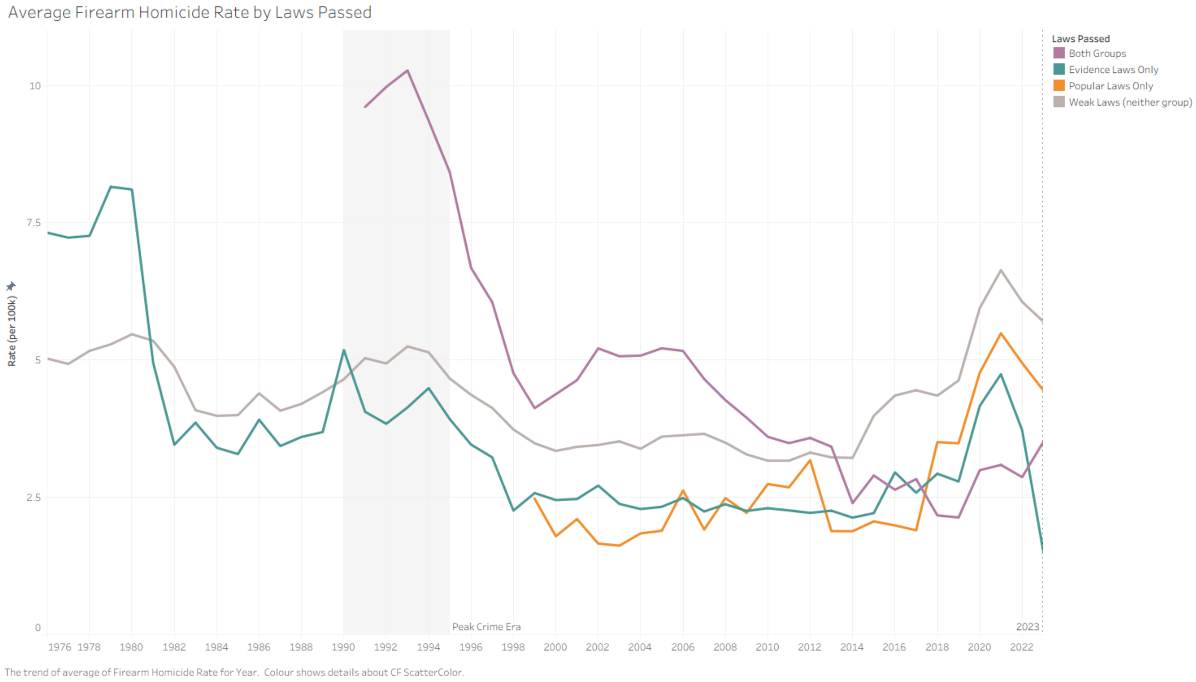

Figure 3: Q2 – Firearm Homicide Rate Over Time by Law Profile (1976–2023)

Insights:

- Higher-law groups show a more pronounced decline in firearm homicide rates from the 1990s onward, and the gap between groups widens over time.
- The pattern holds under both grouping modes (quantity and type).

4.3 Q3 – State-Level Law History and Homicide Standing

Two charts answer Q3[2.6.3]. The first (Figure 4) is a **line chart** showing how the total number of firearm laws in a selected state has changed over time. A **state parameter** (`P State`) drives the selection; this is the second parameter introduced in the project, complementing `P LawCategorization` from Figure 2.

The second chart (Figure 5) is a **bar chart** that ranks all states by firearm homicide rate for a chosen year. The selected state is highlighted using a **set**, and a national average **reference line** is overlaid for context. The reference line feature was first used in Figure 3 and is applied here in the same way. Viewed together, the two charts make it easy to see whether legislative growth in a state coincides with an improvement in its homicide ranking.

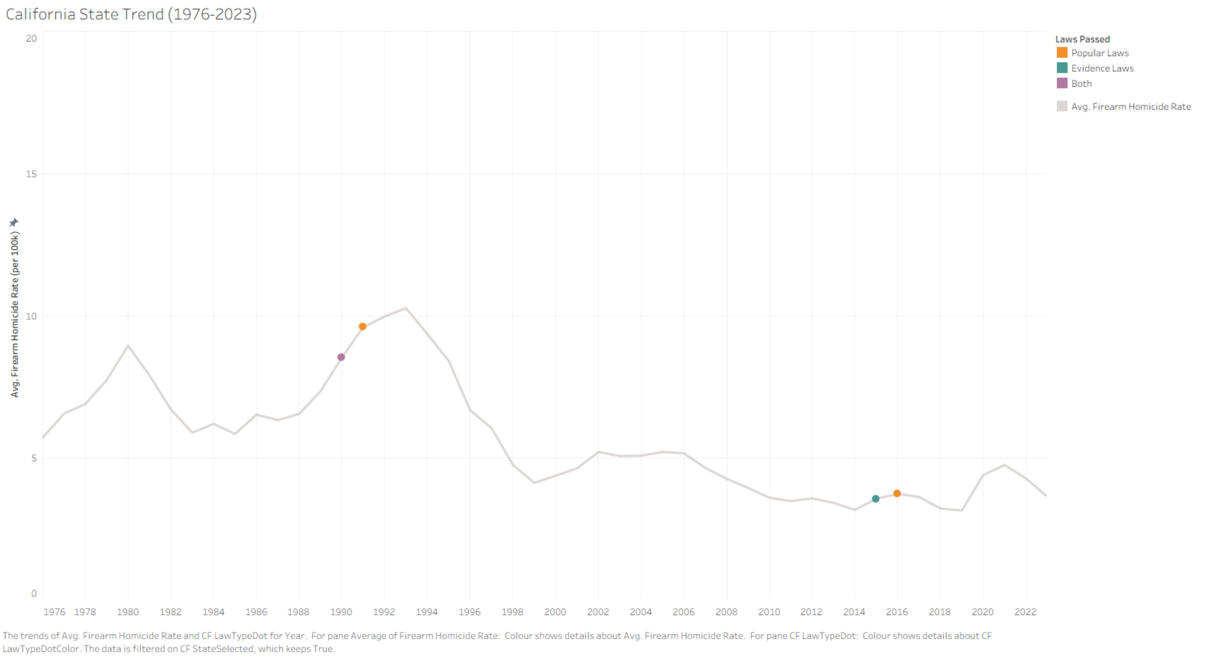

Figure 4: Q3 – Firearm Laws in Effect Over Time for Selected State (Line Chart)

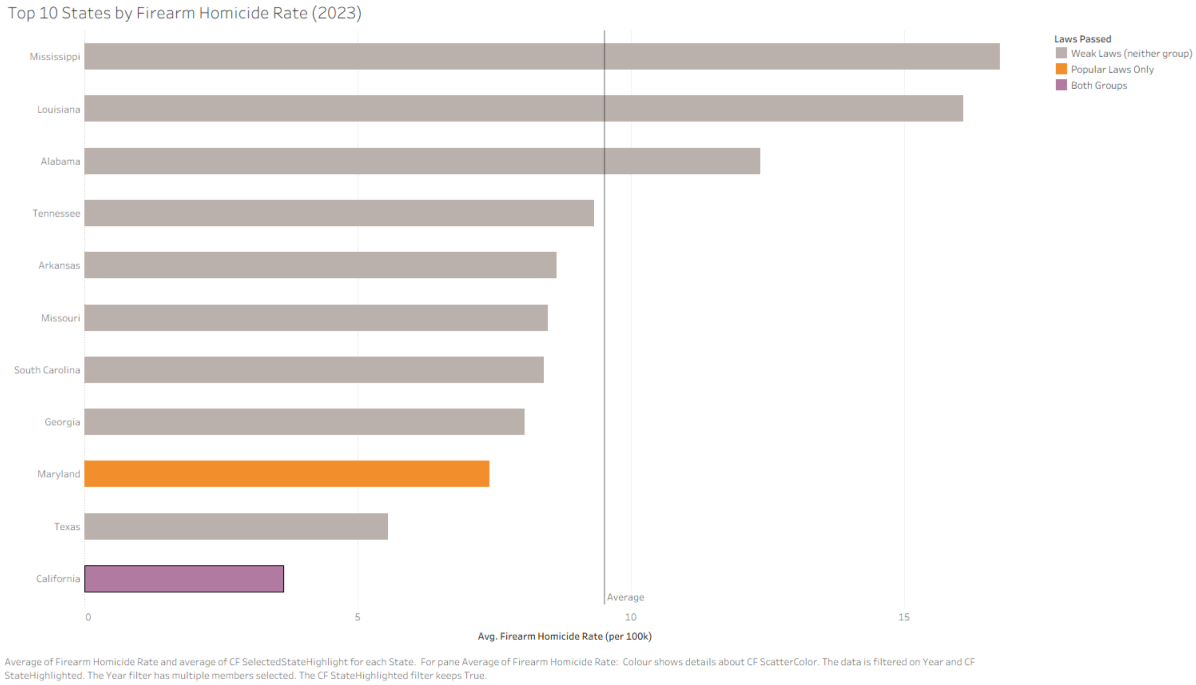

Figure 5: Q3 – Firearm Homicide Rate by State, Selected State Highlighted (Bar Chart)

Insights:

- The bar chart clearly shows that states with more laws and, in particular, more evidence-based laws tend to rank lower in firearm homicide rate compared to other states.
- The line chart makes it difficult to isolate the impact of a single law as soon as it is introduced, since the cumulative law count changes gradually. The aggregate legislative posture of a state is the more legible signal rather than any individual law change.
- In several states, sustained legislative growth aligns with a better position in the homicide ranking, though no causal claim is made.

4.4 Dashboards

Two dashboards bring the individual charts together into a unified interactive experience.

Dashboard 1 (Figure 6) combines the choropleth map (Figure 1), the scatter plot (Figure 2), and the Q2 line chart (Figure 3), covering Q1[2.6.1] and Q2[2.6.2] in one view. Clicking a state on the map triggers a **filter action** that highlights the selected state in the scatter plot. The `P Year` parameter controls the year shown on the map and, simultaneously, drops a **reference line** at that year in the Q2 line chart, so the user can see where a specific year sits relative to the overall trend. The shared `P LawCategorization` parameter applies the same law grouping across all three views at once.

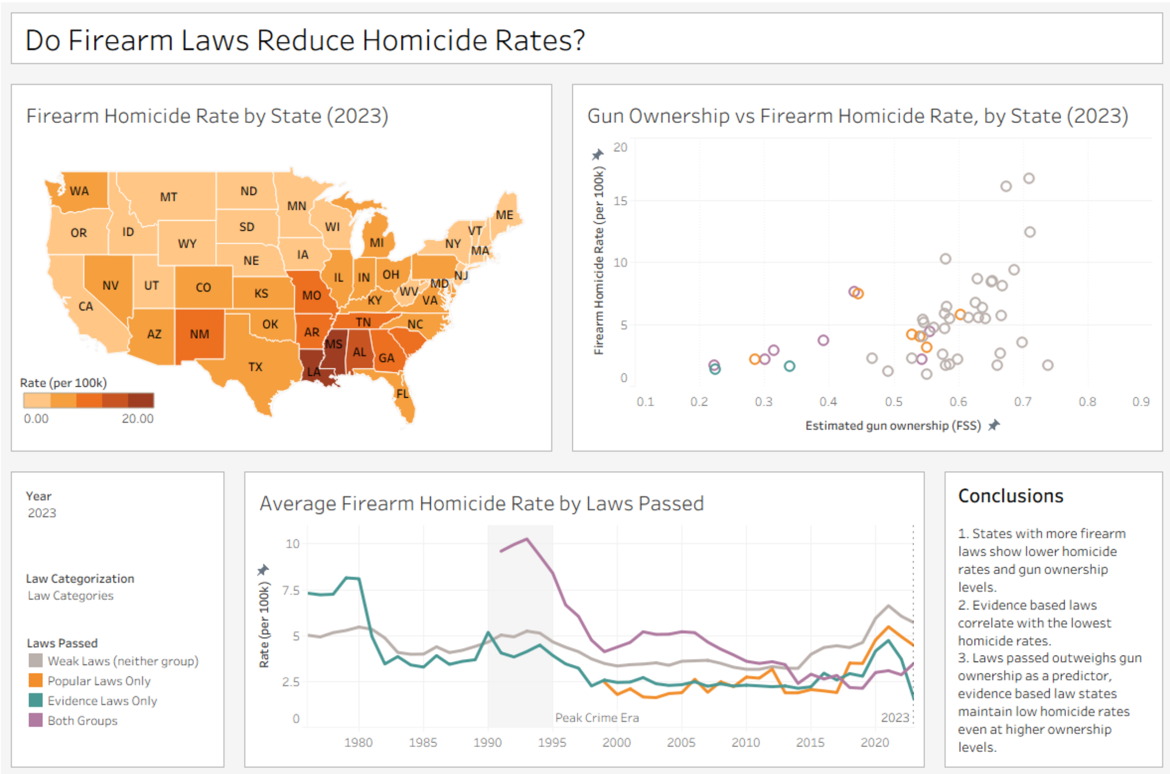

Figure 6: Dashboard 1 – Geographic Overview, Law–Homicide Scatter, and Law Profile Trends (Q1 + Q2)

Dashboard 2 (Figure 7) is built around Q3[2.6.3], with the choropleth map (Figure 1) kept as a geographic reference. The bar chart (Figure 5) and the state law history line chart (Figure 4) are both linked by the shared `P State` parameter introduced in Figure 4, so selecting any state updates both views at once. Clicking a state on the map also triggers a **filter action** that updates the bar chart and the law history line, providing a seamless single-state deep-dive experience.

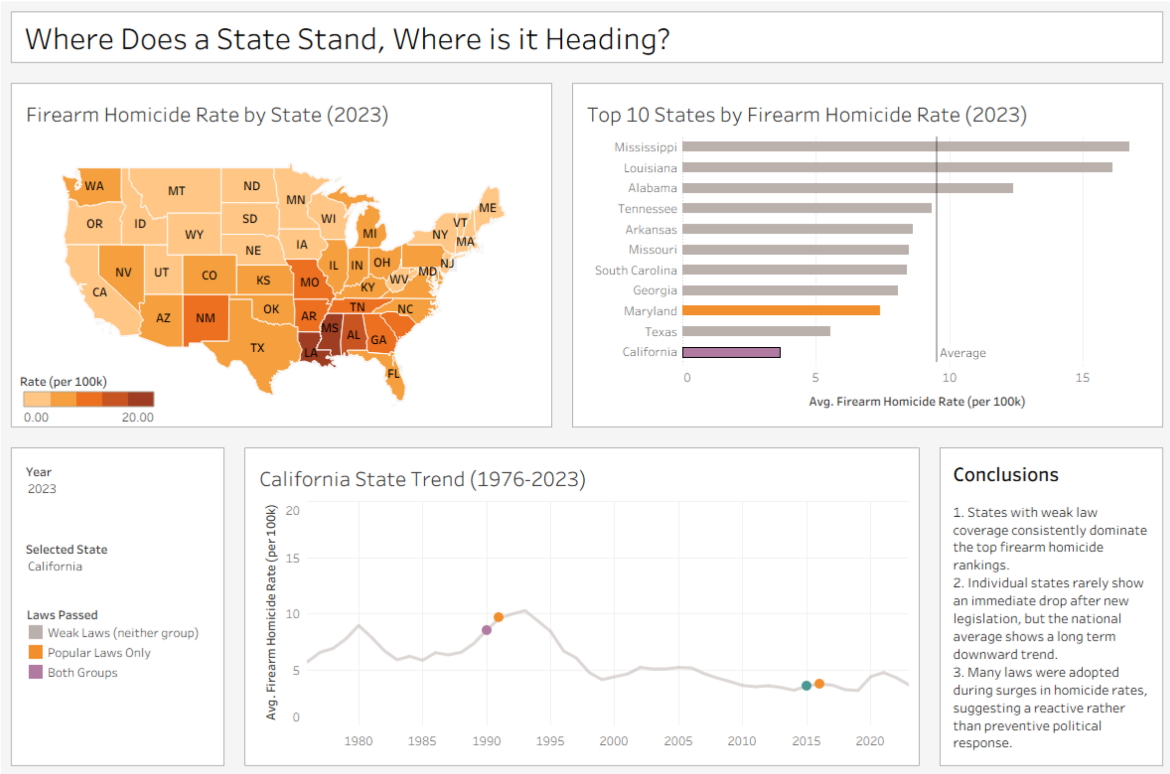

Figure 7: Dashboard 2 – Geographic Map, State Homicide Ranking, and Selected State Law History (Q3 + Q1/Q2)

4.5 Visualization Summary

Q	Chart	Chart Type	Tableau Features
Q1[2.]	Fig. 1	Choropleth map	P Year, CF StateAbbreviation, orange severity gradient
Q1[2.6.1]	Fig. 2	Scatter plot	P LawCategorization (introduced here), color encoding
Q2[2.]	Fig. 3	Line chart (interactive)	P LawCategorization (reused), reference line (introduced here)
Q3[2.6.3]	Fig. 4	Line chart (per state)	P State (introduced here), filter
Q3[2.]	Fig. 5	Bar chart (comparative)	Set, reference line (reused), set action, P State (reused)
Q1+Q2	Fig. 6	Dashboard	Filter action (map to scatter), P Year reference line in line chart, P LawCategorization (reused)
Q3+	Fig. 7	Dashboard	P State (reused), filter action, map click action

Table 2: Visualization summary

Across the project, Table 2 shows:

- Four distinct chart types: choropleth map, scatter plot, line chart, bar chart
- Tableau features used: calculated fields, parameters, sets, dashboard filter and set actions, reference lines, filters, and custom tooltips
- Two data sources joined by state and year, in two different formats (`.csv` and `.xlsx`)

5 Conclusions

Question 1[2.6.1]: Homicide Rates and the Law Landscape

- States with fewer firearm laws tend to have higher homicide rates, visible in both the choropleth map and the scatter plot (Figures 1 and 2).
- The map reveals a clear geographic division: lower-rate states cluster in the north and northeast, while higher-rate states concentrate in the south, correlating with the type and quantity of legislation in each region.

Question 2[2.6.2]: Trends Over Time

- States with more laws, whether by quantity or by type, show a stronger decline in firearm homicide from the 1990s onward (Figure 3).
- The gap between law profile groups widens over time, and the pattern holds under both the quantity-tier and the law-type grouping modes.

Question 3[2.6.3]: State-Level Law History and Ranking

- Although individual law introductions are difficult to track on a timely, state-by-state basis in the line chart, the effects become clearer when ranking states against each other.
- States with more laws and more evidence-based laws consistently rank lower in firearm homicide rates than states with fewer or weaker legislation (Figures 4 and 5).
- Evidence-based laws appear to be a stronger indicator than the raw count of laws: even some states with a high total law count still record elevated homicide rates, suggesting that the *quality*

and type of legislation matters more than its quantity alone.

- That said, more laws still generally outperform fewer laws, and evidence-based laws outperform popular laws when comparing group averages.

Overall, the regulatory environment is the main lens throughout the project. FSS serves as useful background context for estimated ownership, but high ownership alone does not explain the variation in homicide rates: the type and strength of a state's firearm legislation is consistently the more predictive factor. Some limitations apply: FSS is a proxy rather than a direct ownership measure, and the analysis does not control for confounding factors such as poverty, urbanization, or policing levels.

6 Supplementary Analysis: Law Adoption and Homicide Rates Over Time

To complement the Tableau visualizations, an event-study style chart (Figure 8) was generated with AI to show how the average firearm homicide rate evolves around the moment a specific law is adopted. For each law, year 0 is the year of first adoption in a given state; the chart shows the average across all states that adopted the law, from five years before to ten years after adoption.

> **Prompt:** *Generate a multi-line event-study chart showing the average firearm homicide rate from 5 years before to 10 years after the adoption of five key firearm laws, using the Kang et al. and Tufts CTSI datasets. Solid lines for evidence-based laws, dashed for popular laws. Only output the code.*

> **Response summary:** Produced a Plotly multi-line chart with a vertical reference line at year 0, solid colored lines for the three evidence-based laws (Universal Background Checks, GVRO, Assault Restriction) and dashed grey lines for the two popular laws (Concealed Carry Permit, Waiting Period). Direct end-labels are used instead of a legend to reduce clutter.

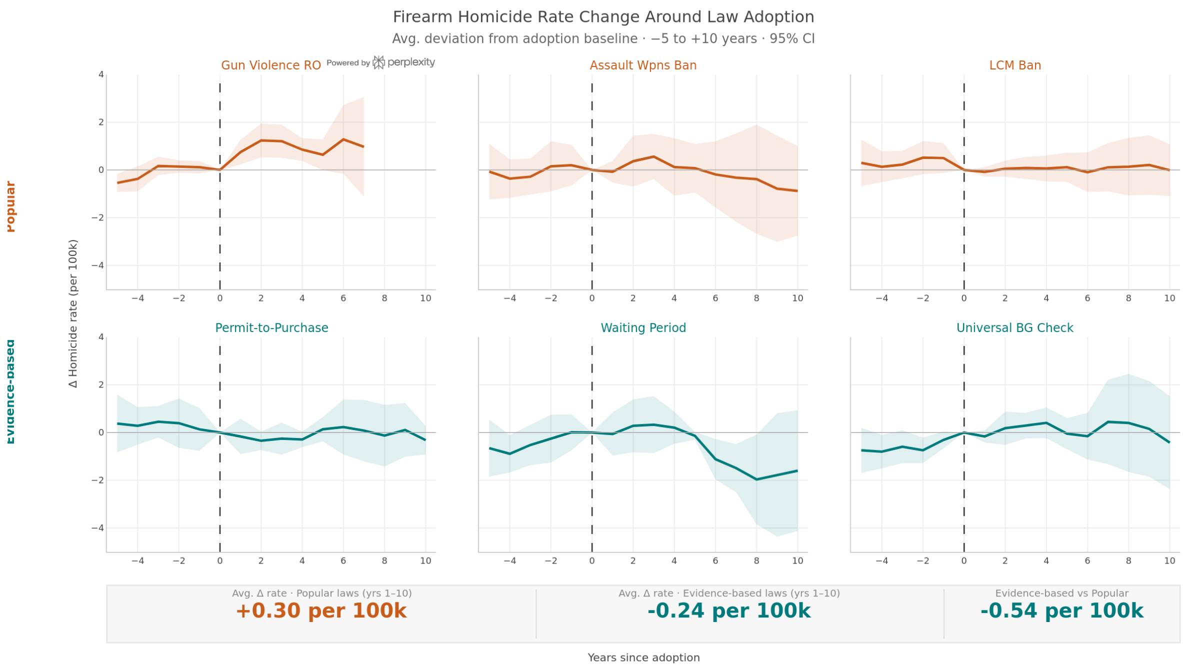

Figure 8: Supplementary – Average Firearm Homicide Rate Around Law Adoption (Event Study, -5 to $+10$ years)

Insights:

- All five laws are associated with a decline in firearm homicide rate in the years following adoption, but the steepness of the decline differs substantially by law type.
- Evidence-based laws (solid lines) show a noticeably steeper and more sustained post-adoption decline than popular laws (dashed lines), consistent with the findings from the Tableau bar chart (Figure 5).
- Universal Background Checks shows the sharpest improvement, followed by GVRO and Assault Restriction.
- Popular laws (Concealed Carry Permit and Waiting Period) show a modest but slower improvement, reinforcing the conclusion that the type of law matters more than its prevalence alone.

7 References

[1] Seokho Kang et al. *Firearm Suicide Proxy for Household Gun Ownership, 1949–2023*. Accessed: 2026-05-18. 2023. DOI: 10.7910/DVN/QVYDUD. URL: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QVYDUD.

[2] State Firearm Law Research Group. *State Firearm Law Database, 1976–2024*. Accessed: 2026-05-18. Tracks presence or absence of 72 firearm law provisions across all 50 U.S. states, 1976–2024. 2024. URL: https://www.tuftsctsi.org/state-firearm-laws/.

[3] Seokho Kang and Taehee Lee. "State-level household gun ownership proxy dataset, 1949–2020". In: *Data in Brief* 50 (2023), p. 109548. DOI: 10.1016/j.dib.2023.109548. URL: https://www.sciencedirect.com/science/article/pii/S2352340923006480.

[4] Seokho Kang et al. "Extending the Firearm Suicide Proxy for Household Gun Ownership". In: *American Journal of Preventive Medicine* (2023). DOI: 10.1016/j.amepre.2023.06.004. URL: https://pubmed.ncbi.nlm.nih.gov/37160211/.

[5] Christopher S. Koper and Jeffrey A. Roth. *Impacts of the 1994 Assault Weapons Ban: 1994–96*. Tech. rep. NCJ 173405. Research in Brief. Congressionally mandated assessment of the Federal Assault Weapons Ban. National Institute of Justice, U.S. Department of Justice, 1999. URL: https://www.ojp.gov/pdffiles1/173405.pdf.

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[8] Tableau Software. *Tableau Desktop*. 2026. URL: https://www.tableau.com/products/desktop.